Ashkan Arabi

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EDUCATION

Bachelor of Science, Computer Science December 2025
The University of Texas at El Paso GPA: 3.93
Certifications: AWS Certified Solutions Architect - Associate (April 2026)

EXPERIENCE

Capital One: Associate Software Engineer February 2026 - present | New York, NY

- Developed a POC semantic complaint routing engine using embeddings and **ChromaDB** cosine similarity to match incoming complaints with product teams, targeting a **40% historical misassignment rate**.
- Implemented an online relevance-feedback algorithm (Rocchio-style feedback) to adjust team embeddings from product manager accept/reject decisions, with **asynchronous Kafka-based reassignment**.
- Built **AWS Lambda Golang microservice**, to orchestrate internal clients' R/W operations on customer tax information at up to **200TPS** while they're gradually migrated from legacy Profile DB to in-house DB.
- Created **Claude Code** skill to automate onboarding tickets into code changes and GitHub PRs, currently **in use across customer orchestration domain with 30+ engineers**.
- Wrote **Grafana K6** load tests for Lambdas in QA env simulating **Discover data migration** onto infrastructure.

Capital One: Software Engineering Intern June 2025 - August 2025 | McLean, VA

- Built & deployed **anomaly-detection pipeline** for Capital One's card decisioning system, notifying SREs of malfunctioning ML models and reducing incident-response time **from 4+ hours to <15 mins**.
- Wrote highly optimized **Databricks SQL** & evaluated warehouse sizes to minimize pipeline cost & latency.
- Extended **Java**-based plugin to log custom **Apache Spark** data lineage events using a YAML specification.

UTEP Vision and Learning Lab: Undergraduate Research Assistant January 2024 - May 2024 | El Paso, TX

- Built tone-changer by fine-tuning **HuggingFace** models to change normal tone to autistic.
- Implemented neural style-transfer model in **PyTorch** to explore usage for accent-changing.
- Increased evaluation script performance by **144x** reimplementing KNN in PyTorch to utilize GPU.

PROJECTS

Movie Rec Engine | FAISS, Redis, gRPC, PyTorch, FastAPI, Postgres, AWS

- Implemented production-grade two-stage (candidate-retrieval + ranking) **collaborative-filtering**-based movie recommender, deployed to **GPU-enabled AWS EC2**.
- Separated retrieval & ranking services for independent scaling, connected by **gRPC** for minimum overhead.
- Used **FAISS** for GPU-accelerated candidate retrieval & **pre-warmed, Redis + Postgres write-through cache** to minimize latency when fetching repeated embeddings.

Ashdocs | Go(Golang), Websocket, Automerge

- Built **multi-threaded** real-time collaborative editor in **Go**, capable of **930+ edits/s** on Apple M4.
- Integrated **Automerge CRDTs** for conflict-free merging of simultaneous edits with **eventual consistency**.
- Designed event pipeline using **goroutines** and **WebSockets** to broadcast per-document updates.
- Created **Pebble**-backed operation log w/ **periodic full snapshots** to cap operation replay time.

Newsbridge (news.ashkan.zone) | FastAPI, React, Postgres, JavaScript, Docker, Ollama, Gemini API

- Increased access to diverse news by creating an AI RSS news translator & orator with **55 GitHub stars**.
- Allowed instant playback using **async** programming to **concurrently generate** summary & play audio.

SKILLS

Languages: Go (Golang), Python, Java, TypeScript/JavaScript, SQL, Bash
Frameworks & Libraries: ReactJS, Spring Boot, PyTorch, TensorFlow, JUnit
Infra & Tools: AWS, Docker, Claude Code, Kafka, Redis, Databricks, New Relic, Git